



NCEA Level 3 Calculus

AS 91577
Complex Numbers

Learning Workbook
2026 Edition

Achievement Standard 91577

Apply the Algebra of Complex Numbers in Solving Problems

Achievement Criteria

Achievement	Achievement with Merit	Achievement with Excellence
Apply the algebra of complex numbers in solving problems.	Apply the algebra of complex numbers, using relational thinking, in solving problems.	Apply the algebra of complex numbers, using extended abstract thinking, in solving problems.

Notes

1. Apply the algebra of complex numbers in solving problems involves:

- selecting and using methods
- demonstrating knowledge of concepts and terms
- communicating using appropriate representations.

Relational thinking involves one or more of:

- selecting and carrying out a logical sequence of steps
- connecting different concepts or representations
- demonstrating understanding of concepts
- forming and using a model;

and also relating findings to a context, or communicating thinking using appropriate mathematical statements.

Extended abstract thinking involves one or more of:

- devising a strategy to investigate or solve a problem
- identifying relevant concepts in context
- developing a chain of logical reasoning, or proof
- forming a generalisation;

and also using correct mathematical statements, or communicating mathematical insight.

2. Problems are situations that provide opportunities to apply knowledge or understanding of mathematical concepts and methods. Situations will be set in real-life or mathematical contexts.
3. Methods are selected from those related to:
 - quadratic and cubic equations with complex roots
 - Argand diagrams
 - polar and rectangular forms
 - manipulation of surds
 - manipulation of complex numbers
 - loci
 - De Moivre's theorem
 - equations of the form $z^n = r\text{cis}\theta$, or $z^n = a + bi$, where a, b are real and n is a positive integer.

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Chapter 1

Quadratic & Polynomial Equations

LEARNING OBJECTIVES

By the end of this chapter, you should be able to:

1. **Solve Surd Equations:** Use algebraic isolation and squaring techniques to solve equations involving square roots, ensuring all solutions are rigorously checked for validity.
2. **Analyse the Discriminant:** Determine the nature of roots for quadratic equations and solve inequalities involving unknown algebraic coefficients.
3. **Perform Polynomial Division:** Execute algebraic long division to find precise quotients and remainders.
4. **Apply Remainder & Factor Theorems:** Utilise substitution to rapidly evaluate remainders and formally prove algebraic factors without long division.
5. **Apply Roots and Coefficients:** Deploy Vieta's Formulas to manipulate and prove properties of quadratic and cubic equations algebraically.

TOPIC 1 · SOLVING SURD EQUATIONS

A surd equation contains an unknown variable inside a square root. To solve these equations, you must systematically eliminate the square root:

1. **Isolate the surd** on one side of the equation.
2. **Square both sides** to eliminate the radical. Ensure you expand binomials correctly: $(a \pm b)^2 = a^2 \pm 2ab + b^2$.
3. **Rearrange** the resulting equation into a standard polynomial (typically a quadratic) and solve for x .

COMMON ERROR

Squaring both sides of an equation can introduce mathematical "ghosts" known as extraneous roots. You **must** substitute your final values of x back into the original equation. If a value results in a mathematical contradiction, it must be explicitly rejected.

Example 1

(NCEA 2014) Solve the equation

$$x = \sqrt{33 - 4x} + 3$$

Solution:

1. **Isolate the surd:**

$$x - 3 = \sqrt{33 - 4x}$$

2. **Square both sides:**

$$(x - 3)^2 = 33 - 4x \implies x^2 - 6x + 9 = 33 - 4x$$

3. **Rearrange into a quadratic and solve:**

$$x^2 - 2x - 24 = 0 \implies (x - 6)(x + 4) = 0$$

The theoretical solutions are $x = 6$ and $x = -4$.

4. **Check for extraneous roots:** Substitute $x = 6$: LHS = 6. RHS = $\sqrt{33 - 24} + 3 = \sqrt{9} + 3 = 6$. (Valid)
Substitute $x = -4$: LHS = -4. RHS = $\sqrt{33 - (-16)} + 3 = \sqrt{49} + 3 = 10$. (Invalid)

5. **Final Answer:** The only valid solution is $x = 6$.

Example 2

(NCEA 2016) Solve the equation $5 - \sqrt{x} = \sqrt{x-p}$ for x in terms of p .

Solution:

1. **Square both sides:**

$$(5 - \sqrt{x})^2 = x - p \implies 25 - 10\sqrt{x} + x = x - p$$

2. **Simplify and isolate the surd:**

$$25 - 10\sqrt{x} = -p \implies 10\sqrt{x} = p + 25 \implies \sqrt{x} = \frac{p + 25}{10}$$

3. **Square both sides again to solve for x :**

$$x = \frac{(p + 25)^2}{100}$$

Example 3

(NCEA 2018) Solve the equation

$$\frac{\sqrt{x+k} + \sqrt{x-k}}{\sqrt{x+k} - \sqrt{x-k}} = 4$$

for x in terms of k .

Solution:

1. **Cross-multiply to remove the fraction:** Let $a = \sqrt{x+k}$ and $b = \sqrt{x-k}$ for algebraic clarity.

$$\frac{a+b}{a-b} = 4 \implies a+b = 4(a-b) \implies a+b = 4a-4b$$

2. **Group like terms:**

$$5b = 3a \implies 5\sqrt{x-k} = 3\sqrt{x+k}$$

3. **Square both sides:**

$$25(x-k) = 9(x+k)$$

4. **Expand and solve for x :**

$$25x - 25k = 9x + 9k \implies 16x = 34k \implies x = \frac{34k}{16} = \frac{17k}{8}$$

Example 4

(NCEA 2020) Solve the following equation for x in terms of g :

$$2\sqrt{x} - 5 = \sqrt{4x - g}$$

Solution:

1. **Square both sides immediately:**

$$(2\sqrt{x} - 5)^2 = 4x - g$$

$$4x - 20\sqrt{x} + 25 = 4x - g$$

2. **Simplify and isolate the remaining surd:** The $4x$ terms cancel out on both sides.

$$-20\sqrt{x} + 25 = -g \implies 20\sqrt{x} = g + 25$$

$$\sqrt{x} = \frac{g + 25}{20}$$

3. **Square both sides again to solve for x :**

$$x = \left(\frac{g + 25}{20}\right)^2 = \frac{(g + 25)^2}{400}$$